Crop Dynamic Root-Depth-Matched High-Resolution Multilayer Soil Moisture Dataset Generated by In Situ-Remote Sensing Data Fusion

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***Abstract*—Frequent agricultural droughts pose a serious threat to crop production. Soil moisture (SM), as a key monitoring indicator, is essential for developing effective agricultural drought indices. Most existing SM-based agricultural drought indices overlook variations in crops' water requirements across developmental stages and root-zone depths, instead focusing on a single SM layer (surface or root zone). A primary challenge for field-scale agricultural drought monitoring is the lack of high-spatial-resolution, multilayer SM datasets. To address this gap, we propose a data-fusion framework that integrates remote sensing products and pointwise in situ observations to generate daily, 30-m resolution, multilayer SM across the root zone. The newly developed multilayer SM dataset demonstrated excellent performance, with an average Pearson correlation coefficient (PCC) of 0.989 and root-mean-square error (RMSE) of 0.034 m^3 . We also tested spatiotemporal generalization and confirmed that the fusion approach robustly captures field-scale SM when downscaling from 9 km to 30 m. Using this multilayer dataset, we developed the dynamic root-depth-matched agricultural drought index (DRDI), which dynamically adapts to crop root-depth distributions across phenological stages. Spatiotemporal analysis using DRDI revealed drought conditions across areas and seasons. Compared with existing indices, DRDI better reflected drought impacts on crop yield and provided more accurate field-scale agricultural drought assessments. Through data sensitivity analysis, DRDI demonstrated resilience to input perturbations. This work has important implications for Geo-AI-based generation of high-quality SM data. More importantly, the proposed drought index offers new perspectives for the geoscience community and agricultural drought applications.**

***Index Terms*—Agricultural drought, crop development, data fusion, multilayer soil moisture (SM).**

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I. Introduction

UNDER global climate change, the increasing frequency of extreme drought events, along with their impacts, has drawn growing attention worldwide [1], [2], [3]. As one of the extreme drought events, agricultural drought refers to the unusual reduction in soil moisture (SM) during the growth period of crops [4], [5], [6]. It is a complex and destructive disaster that has resulted in substantial losses in agricultural production and consequently exerted a significant influence on socio-economic development [7], [8], [9]. Establishing effective agricultural drought monitoring indicators holds great significance for drought mitigation [10].

Generally, in the context of agricultural drought monitoring, crops serve as the principal subjects. The degree of water stress endured by crops is closely correlated with the content of SM. As a result, SM is commonly recognized as a crucial indicator for monitoring agricultural droughts [11], [12]. Nevertheless, a monitoring indicator incorporating solely SM exhibits certain limitations when it comes to drought monitoring in diverse regions, across different seasons, and for various types of droughts. In light of this, multivariate indices have been developed to quantify agricultural drought from multiple aspects

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[13]. For example, the Palmer drought severity index (PDSI), proposed by Palmer in 1965, is one of the earliest drought indices that combines SM and meteorological data [14]. With the availability of SM data, McKee et al. [15] proposed the standardized SM index (SSI), which assesses drought by quantifying the degree of SM anomalies into a standardized value through the standardization of SM values. Subsequently, the agricultural reference index for drought (ARID) proposed by Woli et al. [16] took into account not only SM but also factors such as crop growth requirements and precipitation, allowing for a more accurate reflection of drought conditions in agricultural production. Recent studies have incorporated the existing drought indices to achieve further progress. For example, Varol et al. [17] employed the Copula function to compute the joint probability distribution of the standardized precipitation index (SPI) and SSI, on the basis of which they constructed the multivariate standardized drought index (MSDI). S ánchez et al. [18] integrated the vegetation condition index (VCI), modified temperature condition index (MTCI), and SM condition index (SMCI), and they obtained the SM agricultural drought index (SMADI) by multiplying the SMCI by the MTCI/VCI.

However, existing agricultural drought indices based on SM have seen relatively scant research efforts directed toward the root-zone SM. Instead, the majority of them concentrate merely on the surface SM [19]. It should be noted that the primary means by which crops acquire water is through absorption from the root-zone soil via their roots. Furthermore, variations in root distributions among different crops give rise to disparate water requirements at distinct root-zone depths [20], [21], [22]. In addition, the root conditions of crops differ across various developmental stages, which likewise exerts an impact on the water demand of crops [23], [24], [25]. Regrettably, current research has, to a large extent, failed to adequately consider the dynamic changes that occur throughout the growth and development process of crops. Instead, the overall growth condition of crops is often employed as a proxy for the entire growth and development process [26].

One of the primary challenges is the absence of high-resolution and multilayer SM datasets [27], [28]. For instance, existing global gridded SM products with relatively fine spatial and temporal resolution include the 9-km Soil Moisture Active Passive (SMAP) product and the 0.1° ERA5-Land product. However, such resolutions still remain inadequate for field-scale studies (e.g., only one pixel in ERA5-Land for an area of 100 km^2). Therefore, to establish a root-zone SM-based drought monitoring method, it is crucial to generate high-resolution and multilayer SM datasets by integrating various satellite-based, reanalysis, and in situ observations through advanced downscaling techniques [29], [30], [31].

The primary SM downscaling methods generally integrate microwave or optical/thermal satellite SM products to enhance spatial resolution or high spatial consistency [32], [33], [34]. Another common SM downscaling method involves data assimilation techniques based on global hydrological models (GHMs) or land surface models (LSMs) [35], [36]. In recent years, increasing efforts have been dedicated to downscaling SM by incorporating machine learning approaches and

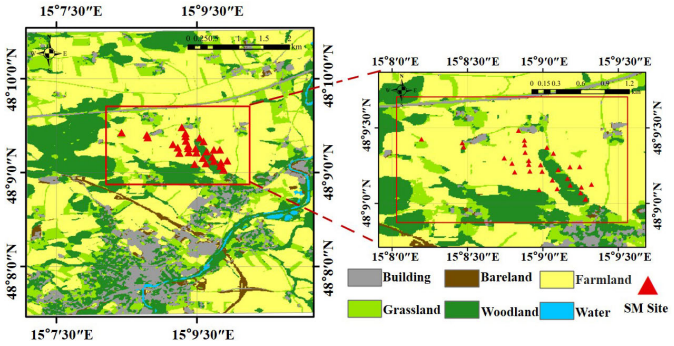


Fig. 1. Geographical location and land cover distribution of the HOAL. The figure displays a total of 31 sites and 6 land cover types.

SM-related variables (e.g., precipitation, temperature, and vegetation indices) [37], [38], [39]. However, most existing studies focus on the satellite-satellite or model-based SM data fusion, which mainly produce large-scale surface SM data, while ground-satellite data fusion and root-zone SM data fusion are still in its infant stage [40], [41], [42]. However, without 30-m field-scale and root-zone SM data, accurate agricultural drought monitoring cannot be fully achieved. To this point, in situ and remote sensing-based SM data fusion makes it promising to generate high-resolution and multilayer SM datasets for field-scale agricultural drought monitoring.

Overall, most existing hybrid and statistical downscaling methods for drought-related data have primarily concentrated on improving the spatial resolution of surface SM. However, these approaches often overlook the vertical heterogeneity of soil water availability in the root zone and fail to account for crop-specific rooting characteristics and developmental dynamics. Against this backdrop, two key gaps remain: 1) the lack of high-resolution, multilayer SM datasets that reflect water availability in the root zone; and 2) the absence of regional drought indices that dynamically incorporate crop-specific root structures and developmental stages.

To overcome these limitations, this study aims to address the following list.

1) Develop an advanced SM fusion framework that integrates satellite, reanalysis, and in situ observations to downscale SM into high-resolution (30 m, daily) and multilayer datasets. This work will address the limitation of surface SM data at coarse resolution.
2) A new drought index, named the dynamic root-depth-matched agricultural drought index (DRDI) by incorporating multilayer SM datasets and crop root information across different root-zone depths and development stages. By jointly addressing the lack of high-resolution multilayer SM data and the absence of root-sensitive drought indices, the proposed approach will substantially improve the drought monitoring capability based on geoscience techniques and remote sensing approaches.

II. Study Area and Data

A. Study Area

The study area is located in Petzenkirchen, Austria, at the geographical coordinates of 48°9′N and 15°9′E, covering

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TABLE I
DESCRIPTION OF THE DATASETS USED IN THIS STUDY

Type	Product	Variable	Spatial Resolution	Temporal Resolution
Remotely sensed data	Landsat 8	Surface reflectance	30 m	16-day
		Thermal infrared	100 m	16-day
	MOD11A1	Land surface temperature	1 km	daily
	MOD09GA	Surface reflectance	500 m	daily
Reanalysis data	SMAP L4	Surface soil moisture (SSM)	9 km	3-hour
		Root zone soil moisture (RZSM)	9 km	3-hour
	ERA5-Land	Soil moisture at layer 1 (0–7 cm), layer 2 (7–28 cm), layer 3 (28–100 cm)	0.1°	hourly
		Precipitation	0.1°	hourly
	SRTM	DEM	30 m	\
In-situ data	ISMN	Soil moisture in layer 1 (0–5 cm), layer 2 (5–10 cm), layer 3 (10–20 cm), layer 4 (20–50 cm)	\	hourly

* The "\" indicates that certain data types do not possess a defined spatial or temporal resolution.

an area of approximately 25 km^2 (Fig. 1). In this region, a SM network of an agricultural catchment, namely, the Hydrological Open Air Laboratory (HOAL), is in operation. This network provides site SM data to the International SM Network (ISMN). Characterized by the densest distribution of SM stations among areas where farmland constitutes the majority of land use, the HOAL serves as a typical experimental region for numerous SM studies at the field scale. The climate within the entire study area is of a warm and humid nature. Over the past two decades, the average annual precipitation was approximately 800 mm, and the average annual temperature was 10 °C. Notably, both temperature and rainfall levels are relatively elevated during the summer season and decline during winter. Farmland dominates over 60% of the total area, with corn as the principal summer crop and wheat as the primary winter crop. Considering local climate patterns, land cover, agricultural traits, and the distribution of SM sensors, the HOAL catchment is an ideal site for the fusion of high-accuracy SM data as well as for the design of agricultural drought indices at the field scale.

B. Data

In this study, three types of datasets were adopted: 1) remotely sensed products, including Landsat 8, MOD11A1, and MOD09GA; 2) reanalysis products, encompassing SMAP Level-4 and ERA-5 reanalysis products; and 3) in situ SM data from the ISMN. Detailed descriptions of these products are given in Table I.

1) Remotely Sensed Datasets: Landsat 8, launched in 2013, represents the eighth satellite in the Landsat series [43]. It carries two key instruments: the Operational Land Imager (OLI) and the Thermal Infrared Sensor (TIRS). The OLI features high spectral resolution and is capable of observing multiple bands, with a spatial resolution of 30 m. The TIRS is mainly utilized for measuring the thermal radiation of the Earth's surface to obtain temperature information, with a resolution of 100 m. We first selected clear-sky images based on cloud information, specifically those with cloud cover less than 5%. Given the 16-day temporal resolution of Landsat 8, we acquired data for the study area at 16-day intervals throughout 2019. Then the thermal infrared bands were resampled to 30 m and 9-km resolution by using cubic convolution and were then applied to retrieve land surface temperature (LST) [44]. In addition, bands (1)–(7) were utilized to calculate NDVI and Albedo. Albedo affects the surface energy balance. An increase in albedo reduces the absorption of solar radiation, thereby suppressing surface evaporation and, to some extent, decreasing SM loss. NDVI reflects vegetation cover density, and vegetation can regulate SM through transpiration. Therefore, albedo and NDVI were selected as input variables for SM downscaling. To obtain the 9-km inputs required for the downscaling model, we also resampled the 30-m LST and NDVI to 9 km.

Meanwhile, we used the daily 500-m MODIS surface reflectance product (MOD09GA) and the daily daytime (local time between 10:00 A.M. and 12:00 P.M.) 1-km MODIS LST product (MOD11A1) for the entire year of 2019. The surface reflectance product supported further calculations of NDVI and Albedo. MOD09GA provides surface reflectance values from Band 1 to 7, which have undergone atmospheric correction (e.g., Rayleigh scattering and gases). For the LST data, we used the associated quality control (QC) information to exclude pixels affected by cloud cover or cloud shadows, retaining only those with good quality. LST is directly related

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to the soil evaporation process—for example, moist areas typically exhibit lower temperatures due to evaporative cooling. As such, LST is an important variable that affects SM changes and supports SM downscaling.

In addition, we also applied a reconstruction algorithm based on the Savitzky–Golay (SG) filter to address the limitation from cloud contamination and atmospheric effects. The SG filter is a technique that performs local polynomial fitting within a moving window, enabling signal smoothing while preserving local features of the original data. It has been widely used for smoothing and gap-filling in time series analysis.

2) Reanalysis Datasets: SMAP satellite mission, operated by the National Aeronautics and Space Administration (NASA), was launched on January 31, 2015 [45]. This study utilized the SMAP L4 data, which provides surface SM at the top 5 cm and root-zone SM at the depth of 0–100 cm. It has a spatial resolution of 9 km and a temporal resolution of 3 h. Unlike the SMAP L3 product, the SMAP L4 combines estimation from NASA's catchment-LSM (CLSM) with assimilated brightness temperature data to enhance SM accuracy. In recent years, multiple peer-reviewed studies have validated the reliability of the SMAP L4 product. In this study, SMAP L4 data in the study area for the year 2019 were obtained. Then, we aggregated the original 3-h data to a daily scale. These datasets were used as the core variables for SM data fusion.

ERA5-Land (ERA5) reanalysis dataset, developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), represents the latest generation of global reanalysis products based on the 4-D-Var data assimilation system [36]. It integrates satellite observations, model-based simulations, and in situ measurements, with a spatial resolution of 0.1° and hourly temporal resolution. In this study, we extracted multilayer SM data (i.e., 0–7 cm, 7–28 cm, and 28–100 cm) and precipitation data for the year 2019 from ERA5-Land, and the hourly data were composited to a daily scale. The ERA5-Land SM data were used as a background field to support the downscaling of SM. Precipitation is the primary source of SM and directly determines the spatiotemporal distribution of SM. Accordingly, ERA5-Land precipitation was included as a principal variable in the SM downscaling process.

Topography also plays a crucial role in the spatial distribution of SM [46]. In this study, elevation data from the Shuttle Radar Topography Mission (SRTM) were obtained, with a spatial resolution of 30 m. Specifically, we used the SRTM v3, which was released in 2013 as a variable for SM fusion. To ensure consistency with other variables, the digital elevation model (DEM) data were resampled accordingly.

3) In Situ Data: ISMN is currently one of the most widely used platforms for collecting and publishing globally distributed in situ SM data [47]. In this study, SM observations from the HOAL network, a high-density SM network from ISMN, were employed as the primary source of ground-based measurements. The HOAL network is equipped with sensors that provide hourly SM at depths of 5, 10, 20, and 50 cm. In this study, we used the hourly multilayer SM measurements from all 31 sites within the study area throughout the entire year of 2019. Based on the quality flags provided by the ISMN, only SM values labeled as "G" (good) were retained. Finally, the filtered hourly measurements were averaged to obtain daily SM data, in order to match the temporal resolution of other datasets.

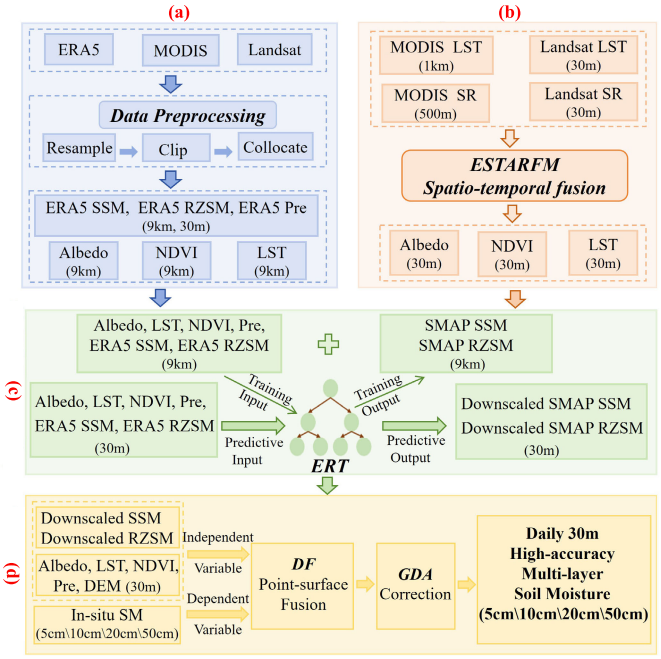


Fig. 2. Framework of the SM fusion. The framework is divided into four parts, including (a) data preprocessing, (b) spatiotemporal fusion, (c) SM downscaling, and (d) SM point-surface fusion.

III. Methods

A. Data Fusion Method for Generating 30-m Resolution and Multilayer SM Datasets

To obtain high-accuracy SM, we propose a new data fusion framework that generates a multilayer SM dataset with a 30-m resolution at the field scale by integrating both surface-surface fusion and point-surface fusion methods (Fig. 2). The surface–surface fusion refers to the SM downscaling based on surface datasets. The point–surface fusion refers to the subsequent calibration step, in which in situ observations are incorporated to further correct and refine the downscaled SM. The detailed information of the fusion is described below.

1) Step 1 (Spatiotemporal Fusion Using ESTARFM): SM and related environmental variables (such as surface reflectance, LST, and vegetation index) are generally restricted in terms of temporal or spatial resolution. This aims to furnish spatiotemporally continuous input data (NDVI, LST, and Albedo) with a 30-m resolution for the subsequent SM fusion model. The Enhanced Spatial and Temporal Adaptive Reflectance Fusion Model (ESTARFM) was adopted to downscale the variables related to SM. ESTARFM enhances the spatial resolution by establishing a quantitative model relationship between pixels of low resolution and those of high resolution. It accounts for the temporal trend of reflectance variations and integrates spatial and spectral similarities to reconstruct each central pixel within the search window,

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thus augmenting the precision of the fusion [48]. The core equations of ESTARFM are as follows:

$$F_k\left(x_{w/2}, y_{w/2}, t_p, B\right) = H\left(x_{\frac{w}{2}}, y_{\frac{w}{2}}, t_k\right) + \sum_{i=1}^{N} W_i * V_i * \left(L(x_i, y_i, t_p, B) - L(x_i, y_i, t_k, B)\right) \quad (1)$$

$$F_k\left(x_{w/2}, y_{w/2}, t_p, B\right) = T_m * F_m(x_{w/2}, y_{w/2}, t_p, B) + T_n * F_n(x_{w/2}, y_{w/2}, t_p, B) \quad (2)$$

where $(x_{w/2}, y_{w/2})$ is the central pixel in the search window of size w $(k = m, n)$. In addition, t_k $(k = m, n)$ is the calculation date, while t_p means the prediction date. N is the number of similar pixels, while W_i and V_i indicate the weight and conversion coefficient of similar pixel i, respectively. B means the bands. H and L represent high-resolution images (HRIs) and coarse resolution images (CRIs), while F is the predicted HRI. Given there are two predicted images t_m and t_n, the predicted fine image is finally the sum of their weighted. The weight at T_k is calculated as

$$T_k = \frac{1 \Big/ \left|\sum_{j=1}^{w}\sum_{i=1}^{w} L\left(x_i, y_j, t_k, B\right) - \sum_{j=1}^{w}\sum_{i=1}^{w} L(x_i, y_j, t_p, B)\right|}{\sum_{k=m,n}\left(1 \Big/ \left|\sum_{j=1}^{w}\sum_{i=1}^{w} L\left(x_i, y_j, t_k, B\right) - \sum_{j=1}^{w}\sum_{i=1}^{w} L(x_i, y_j, t_p, B)\right|\right)} \quad (3)$$

where (x_i, y_i) is the location of ith similar pixel.

2) Step 2 (Two-Layer SM Downscaling Using ERT): SMAP L4 product, as the core variable for SM fusion, provides the surface SM and root-zone SM with high temporal resolution. However, a substantial spatial representation disparity still exists between the large-scale satellite spatial coverage of SMAP L4 (9 km) and the point-scale in situ measurements (approximately 2 m). The scale mismatch issue brings considerable uncertainties to the accuracy of SM fusion [49]. Therefore, to address this issue, the original coarse-resolution input was downscaled to 30 m to reduce scale mismatch errors. We trained the downscaling model at the SMAP L4 scale (9 km) by aggregating all predictors to 9 km, then applied the trained model to 30-m predictors to produce fine-scale predictions. This workflow ensures consistency between the training and prediction phases while enabling the transfer of learned relationships across spatial scales [50], [51]. In particular, we employed the extremely randomized trees (ERTs) algorithm, a tree-based ensemble learning model, as the downscaling model owing to its computational efficiency and excellent capability in processing large datasets. In the process of constructing each individual decision tree, the ERT adopts the strategies of random feature selection and random split points. Through such an approach, enhanced performance can be achieved by incorporating greater randomness and diminishing computational costs [52]. In this case, the 9-km SMAP L4 product was used as the target, while SM background fields ERA5-Land SM, meteorological forcing LST, precipitation, NDVI, and Albedo data were treated as inputs. The model can be summarized as follows:

$$D - SM = f(ERA5SM, LST, Pre, NDVI, Albedo). \quad (4)$$

The $D - SM$ represents downscaled SMAP SSM or RZSM. First, the original environmental variables (ERA5-Land SM, LST, precipitation, NDVI, and Albedo) were aggregated to 9-km resolution to align with the SMAP L4 data using bilinear interpolation prior to training the model. 90% of the pixels were selected as the training set, and the remaining pixels were regarded as the validation set. After the model training, we input downscaled 30-m environmental variables datasets into the best model and finally generated 30-m downscaled SSM and RZSM data.

3) Step 3 (Multilayer SM Point-Surface Fusion Using DF): In this step, we implemented point-surface fusion by integrating in situ measurements to calibrate the downscaled SM data obtained in Step 2. Here, we utilized deep forest (DF) as the fusion model due to its low computational complexity and powerful strength with few data samples and CPU computational resources, as confirmed by prior studies [53]. The model can be summarized as follows:

$$F - SM = f\left(D - SM, LST, Pre, NDVI, Albedo, DEM\right). \quad (5)$$

The $F - SM$ represents fused SM with multiple layers. Compared to the model inputs in (4), the ERA5-Land data was replaced with downscaled SMAP L4 data in Step 2. In addition, DEM was added into the model. The multiple layers (5, 10, 20, and 50 cm) of in situ SM measurements were used as the output of the model.

To further enhance the accuracy of the fused SM, we introduced Geographical Difference Analysis (GDA) as a calibration strategy. The detailed equation of GDA is presented below

$$P_c(x) = P_f(x) \times \sum_{i=1}^{n} \lambda_i (P_{obs}(x_i) - P_f(x)) \quad (6)$$

where $P_{corrected}(x)$, $P_{fusion}(x)$ and $P_{obs}(x_i)$ represent the calibrated SM, the fused SM, and the in situ SM at location x, respectively. λ_i denotes the weight of station i, computed using inverse distance weighting (IDW) based on distance to x. The difference between in situ and fused SM values $P_{obs}(x_i) - P_{fusion}(x)$ is interpolated across the study area using IDW based on the available ground observation sites.

B. Development of the DRDI

The DRDI is proposed as a new agricultural drought index that utilizes the above-generated high-quality multilayer SM with vegetation condition, temperature stress, and crop root distribution information. The DRDI considers diverse water demand of crop roots during the whole phenology, thereby providing a more crop-aware representation of agricultural drought processes.

1) Constituent Indices in DRDI: The first step is to obtain the constituent indices for constructing DRDI, including the VCI, MTCI, root-zone SMCI (RZSMCI), and root length density (RLD).

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TABLE II
RLD OF SUMMER CORN AT DIFFERENT DEVELOPMENT STAGES AND ROOT DEPTHS. RLD REPRESENTS ROOT LENGTH DENSITY

Jointing stage (May - Jun.)		Silking stage (Jul. - Aug.)		Maturity stage (Sept. - Oct.)	
Root Depth /cm	RLD /cm cm^{-3}	Root Depth /cm	RLD /cm cm^{-3}	Root Depth /cm	RLD /cm cm^{-3}
5	10	5	30	5	22
10	18	10	40	10	44
20	28	20	70	20	67
50	15	50	36	50	33

TABLE III
RLD OF WINTER WHEAT AT DIFFERENT DEVELOPMENT STAGES AND ROOT DEPTHS. RLD REPRESENTS ROOT LENGTH DENSITY

Growing stage (Nov. - Mar.)		Flowering stage (Apr.)		Maturity stage (May - Jun.)	
Root Depth /cm	RLD /cm cm^{-3}	Root Depth /cm	RLD /cm cm^{-3}	Root Depth /cm	RLD /cm cm^{-3}
5	13	5	15	5	10
10	13	10	18	10	25
20	25	20	52.	20	60
50	15	50	60	50	75

The VCI assesses the health status of surface vegetation by comparing the differences between the vegetation state in the current period and the most unfavorable as well as the most favorable states in historical records. The MTCI reflects the connection between temperature and vegetation and the excessive wetness induced by temperature via contrasting the differences between the surface temperature in the current period and the lowest and highest surface temperatures in historical records. The RZSMCI is the normalization of the maximum and minimum values of SM in the root zone, which reflects the drought and wetness conditions of SM at different depths. The DRDI is obtained by combining these indices. The calculation formulas for VCI, MTCI, and RZSMCI are as follows:

$$VCI = \frac{(NDVI_i - NDVI_{min})}{(NDVI_{max} - NDVI_{min})} \tag{7}$$

$$MTCI = \frac{(LST_i - LST_{min})}{(LST_{max} - LST_{min})} \tag{8}$$

$$RZSMCI = \frac{(RZSM_{max} - RZSM_i)}{(RZSM_{max} - RZSM_{min})} \tag{9}$$

where $NDVI_i, LST_i$, and $RZSM_i$ are the NDVI, LST, and RZSM values on the ith day, and the $NDVI_{min}$, $LST_{min}, RZSM_{min}, NDVI_{max}$, LST_{max}, and $RZSM_{max}$ are the minimum and maximum values in the historical records, respectively. The value ranges of VCI, MTCI, and RZSMCI are all between 0 and 1. Specifically, a value of 0 for VCI implies an extremely unfavorable vegetation state, a value of 0 for MTCI indicates no temperature stress and a wet state, and a value of 0 for RZSMCI represents a wet condition. Conversely, a value of 1 for VCI indicates an optimal vegetation state, a value of 1 for MTCI implies extremely hot conditions, and a value of 1 for RZSMCI signifies a severely dry state.

The RLD has been newly introduced in this study given that the root states of different crops vary during different developmental stages and at different root-zone depths. The RLD presents a weighting factor to build a physiologically meaningful agricultural drought index, compared with other existing indices. Specifically, four soil depths (5, 10, 20, and 50 cm) were selected. Considering the distinct root distribution characteristics of different crops at various growth stages, corresponding RLD values were assigned to each depth. As field measurement of RLD is challenging, field investigations of root distribution are extremely difficult and costly. Significant efforts have previously been devoted to obtaining the spatial and temporal distribution of RLD for various crops using both experimental measurements and simulations. A substantial body of published data on RLD distribution for different crops exists, providing crucial support for the RLD data sources used in this study [54], [55], [56], [57], [58]. Considering that the two main crops of this study, summer corn and winter wheat, exhibit significant variations in root distribution across their growth stages and vertical soil layers, we extracted RLD data for both crops at three crucial developmental stages and four soil depths from published literature, as presented in Table II (summer corn) and Table III (winter wheat), respectively.

2) DRDI Model: Crops mainly obtain water through absorbing it from the root-zone soil via their roots. Meanwhile, the different root distributions of diverse crops at different development stages also have an impact on SM absorption. This study introduced the SM demands of different crops at different root-zone depths during different development stages to develop an improved SMADI called the soil moisture agricultural drought index for crop development (DRDI). The calculation formula for DRDI is as follows:

$$\begin{aligned} DRDI = &\frac{RZSMCI_{5\,\mathrm{cm}} \times MTCI}{VCI} \times \frac{RLD_{5\,\mathrm{cm}}}{RLD_{all}} \\ &+ \frac{RZSMCI_{10\,\mathrm{cm}} \times MTCI}{VCI} \times \frac{RLD_{10\,\mathrm{cm}}}{RLD_{all}} \\ &+ \frac{RZSMCI_{20\,\mathrm{cm}} \times MTCI}{VCI} \times \frac{RLD_{20\,\mathrm{cm}}}{RLD_{all}} \\ &+ \frac{RZSMCI_{50\,\mathrm{cm}} \times MTCI}{VCI} \times \frac{RLD_{50\,\mathrm{cm}}}{RLD_{all}} \end{aligned} \tag{10}$$

where RLD_{all} represents the RLD of the crop root at a depth ranging from 0 to 50 cm. The ratio of the RLD at each depth to the total RLD is multiplied by the SMADI value for that layer, and finally summed to derive the DRDI. Theoretically, the value range of DRDI is bounded between 0 and 1, where 0 indicates wet conditions, and 1 represents extreme drought stress under active vegetation states. Although VCI is theoretically bounded between 0 and 1, extremely low VCI values mainly occur during dormant or sparsely vegetated periods, during which vegetation indices are dominated by noise rather than physiological signals. To ensure numerical stability, a minimum threshold of 0.05 is imposed on VCI during DRDI computation, such that VCI values smaller than 0.05 are clipped to 0.05. This stabilization strategy prevents

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artificial amplification of drought signals while preserving the physical interpretability of DRDI during vegetated periods.

In the DRDI formulation, the ratio of RLD at each soil depth to the total RLD within the 0–50 cm profile serves as a normalized weighting factor, which is multiplied by the corresponding SM stress term and then aggregated across layers. While formally a weighting procedure, this is not a simple empirical adjustment but is grounded in the biological mechanism of dynamic variation in root water uptake depth. Its physical significance lies in amplifying the relative importance of different soil layers in crop drought resistance. The physiological rationale for this approach is that the crop root distribution across depths determines actual water extraction capacity from different soil layers. DRDI incorporates crop physiological structural information (i.e., root distribution), enabling the index to reflect not only SM status but also the crop's capacity to access water.

To further ensure robustness and avoid numerical instability under low vegetation activity, DRDI is primarily interpreted during active growing seasons, while periods with extremely low vegetation cover are treated separately or masked in agricultural drought assessment. Consequently, DRDI exhibits enhanced biometeorological coupling and improved relevance for agricultural drought identification across different crop types and growth stages.

3) Comparison With Other Agricultural Drought Indices: The SMADI, serving as the basis of the proposed DRDI in this study, is utilized for the evaluation in this study. SMADI is an agricultural drought index that considers the vegetation, temperature, and mere surface SM. The formula is as follows:

$$SMADI = SMCI \times \frac{MTCI}{VCI}. \tag{11}$$

The soil water deficit index (SWDI) considers the decrease or change in water volume, which is calculated using the SM data at different depths obtained through in situ measurements [59]. Since its data source is independent of the DRDI in this study, enabling comparative evaluation. Its formula is as follows:

$$SWDI = \left(\frac{RZSM\text{-}RZSM_{FC}}{RZSM_{AWC}}\right) 10 \tag{12}$$

where $RZSM_{\mathrm{FC}}$ represents the field capacity, which indicates the maximum water content that the soil can hold. $RZSM_{\mathrm{AWC}}$ refers to the available water content of the soil, denoting the amount of water that the soil can provide for crops to utilize. $RZSM_{\mathrm{AWC}}$ is the difference between $RZSM_{\mathrm{FC}}$ and the wilting point $RZSM_{\mathrm{wp}}$ which is calculated based on the hydraulic characteristic curve of the soil column sampled at each depth where the soil water content is measured.

For the three types of agricultural drought indices used for comparative evaluation, we have defined five categories to characterize the state of agricultural soil drought as shown in Table IV.

TABLE IV

CLASSIFICATION OF AGRICULTURAL DROUGHT STATUS BASED ON THE VALUE RANGE OF THE AGRICULTURAL DROUGHT INDEX

SWDI	SMADI	DRDI	Status
> 0	0 ~ 0.19	0 ~ 0.19	No drought
0 ~ -1.99	0.2 ~ 0.39	0.2 ~ 0.39	Slight drought
-2 ~ -4.99	0.4 ~ 0.59	0.4 ~ 0.59	Moderate drought
-5 ~ -9.99	0.6 ~ 0.79	0.6 ~ 0.79	Severe drought
< -10	0.8 ~ 1	0.8 ~ 1	Extreme drought

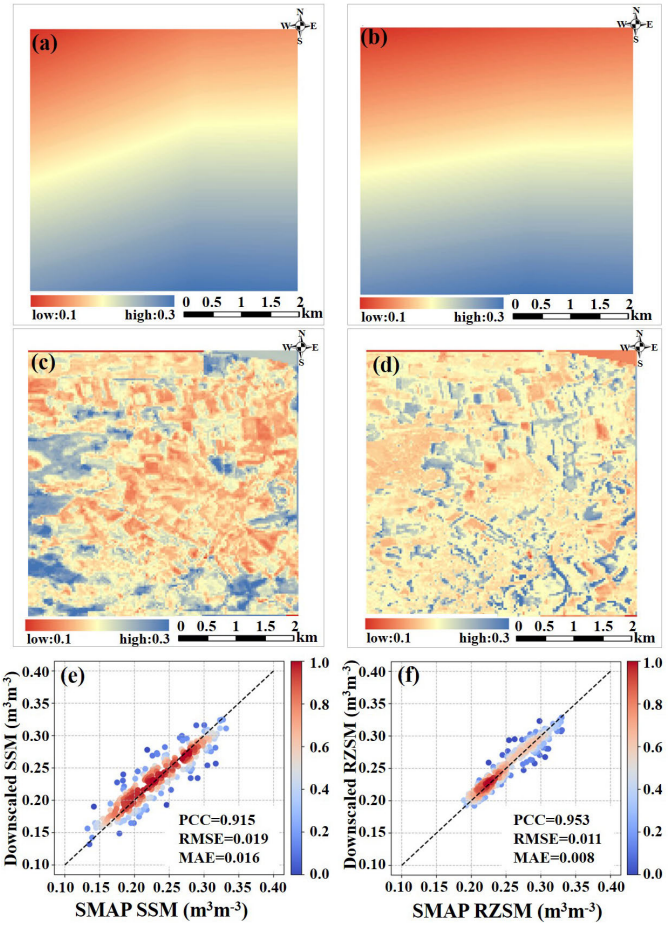


Fig. 3. Spatial consistency between the original SMAP SM and the downscaled product. (a) and (b) Spatial distributions of SMAP SSM and RZSM. (c) and (d) Spatial distributions of downscaled SSM and RZSM. (e) and (f) Scatter density plots illustrating structural correspondence between SMAP and downscaled product.

IV. RESULTS

A. Evaluation of the Fusion SM Dataset

As presented in Section III, a point-surface fusion method was constructed to generate multilayer (5, 10, 20, and 50 cm) daily SM data with a high resolution of 30 m at the field scale. In the first step, we downscaled the original 9-km SMAP surface and root-zone SM products to 30 m using the ERT model. The downscaling results were first examined by comparing the downscaled SM at 30 m with the original SMAP product at 9 km, with the purpose of evaluating spatial consistency across scales. It should be noted that this comparison was not intended as validation, since the downscaled SM is derived from SMAP data. Instead, the aim is to evaluate whether the downscaling method preserves the spatial distribution characteristics of the original SMAP product.

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Here, we have mapped the spatial distribution of the downscaled SM on March 22, 2019, along with the scatter density plot between the downscaled SM and the original SMAP SM (Fig. 3). Due to its coarse spatial resolution, the original SMAP was resampled to 30 m for visualization, without introducing additional spatial information. Fig. 3(a) and (b) shows the spatial distributions of the original SMAP SSM and RZSM, respectively, and Fig. 3(c) and (d) shows the spatial distributions of the downscaled SSM and RZSM, respectively. It was found that the downscaled SM captures enhanced spatial details while maintaining the overall distribution patterns. From the spatial distribution, higher SSM values are manifested in the southwest of the study area, while lower SSM values are presented in the central part. This can be explained by the fact that forests are mainly concentrated in the southwest region, where high NDVI in vegetated areas usually dominates and humid conditions typically prevail. The central and western regions are mainly composed of farmlands and grasslands in the growing stage, and thus the climate there is dry with relatively low SSM values. The spatial distribution of RZSM is similar to that of SSM, but its overall values are relatively higher, which stems from the water storage function of the root-zone soil. Regarding the phenomenon that SSM values are higher than RZSM in the western part, localized deviations may occur under short-term surface processes. In the HOAL, surface SM can increase rapidly following precipitation or irrigation events, while deeper soil layers respond more slowly due to infiltration processes and soil hydraulic properties. This temporal lag may lead to higher SSM than RZSM in certain areas.

Fig. 3(e) and (f) displays the scatter density plots between the downscaled SM and the original SMAP SM. The scatter plots were generated based on a scale-consistent comparison. The downscaled 30-m SM were spatially aggregated to the 9-km resolution to match the spatial support of the SMAP product. Subsequently, the aggregated values were paired with the corresponding SMAP pixels to construct the scatter plots. This indicates a strong positive correspondence between the downscaled SM and the original SMAP SM, reflecting preserved spatial structures across scales. The Pearson correlation coefficient (PCC), root-mean-square error (RMSE), and mean absolute error (MAE) are reported to quantitatively characterize the degree of spatial consistency between the downscaled and original SMAP products. These results confirmed that the ERT model performed effectively in maintaining the spatial structure of the original SMAP product while enhancing spatial resolution.

The validation of the downscaled SSM and RZSM results using the in situ measurements was conducted at the network level. Fig. 4 presents scatter density plots comparing both the original SMAP and the downscaled SM with in situ measurements, where in situ SM serves as the reference for accuracy evaluation. Specifically, the values of SMAP SM and downscaled SM were extracted at the corresponding locations of in situ stations for comparison, and the objective of this comparison is to assess the relative performance between the original SMAP and the downscaled results under a consistent validation framework. The results show that, for both SSM

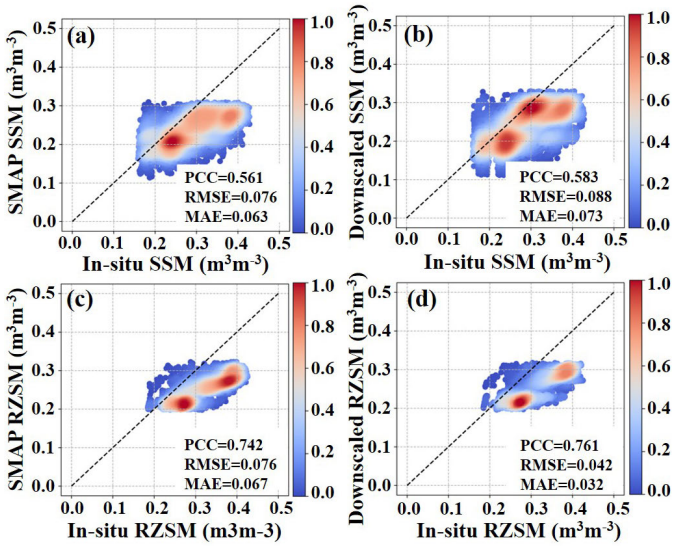


Fig. 4. Scatter plots comparing SMAP SM and downscaled SM with in situ SM. (a) and (b) Scatter plots between SMAP SSM and downscaled SSM with in situ SSM. (c) and (d) Scatter plots between SMAP RZSM and downscaled RZSM with in situ RZSM. The color gradient represents point density, with red colors indicating higher densities of points and blue colors indicating lower densities of points.

and RZSM, higher PCC values and lower RMSE values are observed for the downscaled SM compared with the original SMAP product. This indicates that the downscaled SM exhibits higher correlation and lower error relative to in situ observations. This relative improvement demonstrated that the ERT downscaling approach alleviated the problem of inconsistent spatial matching between 9-km SMAP data and station data. Nevertheless, residual discrepancies between the downscaled SM and in situ observations remain, motivating the subsequent point–surface fusion for further bias correction.

To further enhance the accuracy of the downscaled SM, we conducted point-surface fusion using local observations via the DF model. Fig. 5 displays the spatial distribution of the fused SM and the scatter plot with the in situ SM. Fig. 5(a)–(d) displays the spatial distributions of the fused SM at depths of 5, 10, 20, and 50 cm, respectively. Due to the 9-km coarse spatial resolution of the core variable SMAP for SM fusion, there are only four pixels within the scope of the study area HOAL. Consequently, the fused SM also exhibits a pattern of being divided into four blocks. This segmentation phenomenon can be reasonably explained by the large size of the SMAP pixels. It can be seen that the southwestern region with denser vegetation has relatively higher SM values, while much lower SM values mainly exist in the central part where the vegetation is sparse and less diverse. Moreover, in general, as the soil depth increases, the value of SM also gently increases.

In addition, for consistency in evaluation, the multilayer SM fused by the DF model and the ground in situ measurement data were also validated at the network level. The scatter density plots between the fusion multilayer SM and the in situ SM are shown in Fig. 5(e)–(h). The PCC at different depths of 5, 10, 20, and 50 cm has all surpassed 0.981. Concurrently, the RMSE has not exceeded 0.012 m^3. In

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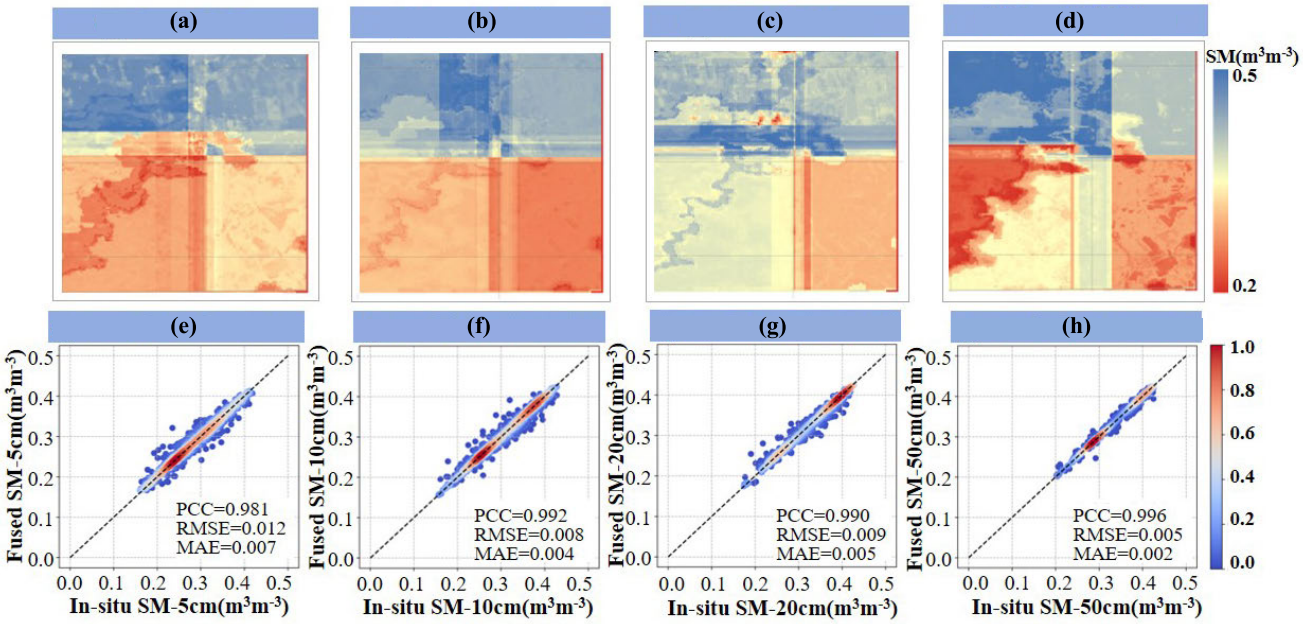


Fig. 5. Spatial distribution and scatter plots of fused SM. The first row shows the spatial distribution of fused SM at depths of 5, 10, 20, and 50 cm. The second row shows the scatter plots of fused SM and in situ SM at depths, respectively. (a) Fused SM-5 cm. (b) Fused SM-10 cm. (c) Fused SM-20 cm. (d) Fused SM-50 cm. (e)–(h) Fused SM versus in situ SM.

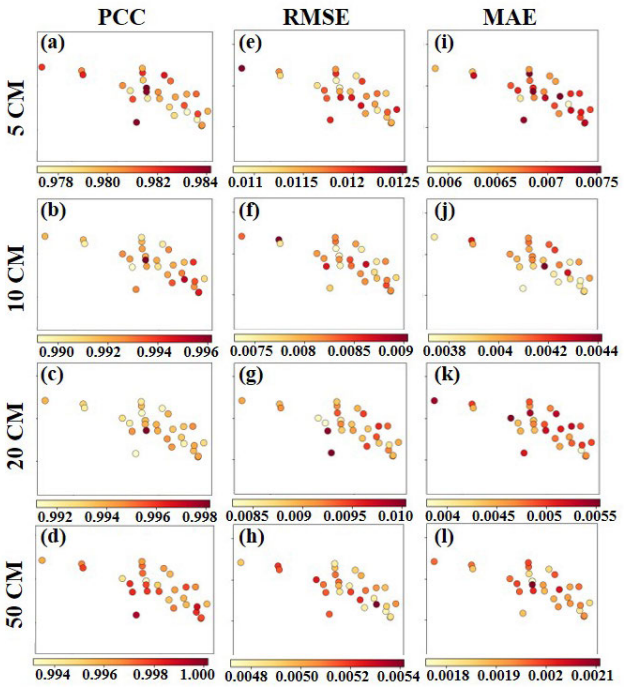


Fig. 6. Spatial distribution of station-level accuracy metrics for the fused multilayer SM. (a)–(d) Spatial distribution of the PCC between the fused SM and in situ observations at depths of 5, 10, 20, and 50 cm, respectively. (e)–(h) Corresponding RMSE. (i)–(l) MAE at the same depths.

general, the PCC is significantly higher, and the RMSE is obviously lower than that of the SM with only downscaling. This result clearly demonstrates the feasibility of employing the DF model for point-surface fusion to improve the accuracy of the downscaled SM.

Complementing the network-level validation, Fig. 6 presents station-wise distributions of accuracy metrics at four soil depths (5, 10, 20, and 50 cm) across all stations in the HOAL to reveal spatial variability in performance. Specifically, Fig. 6(a)–(d) illustrates the spatial distribution of the PCC between the fused SM and in situ observations at all depths, respectively. The corresponding spatial patterns of RMSE and MAE are shown in Fig. 6(e)–(h) and (i)–(l).

A clear improvement with increasing soil depth is observed, with PCC values rising from approximately 0.98 at 5 cm to nearly 1.00 at 50 cm. Besides, slight spatial heterogeneity in model performance can be observed among stations, with stations in the western sector generally showing slightly higher PCC values compared to those in the eastern region across all depths, which is mainly attributed to local heterogeneity in soil properties and vegetation conditions. In terms of error metrics, both RMSE and MAE display coherent spatial patterns across depths and increase systematically with depth. Notably, stations with comparatively higher errors are spatially consistent across depths, suggesting that residual uncertainties are more likely driven by site-specific factors rather than depth-dependent fusion deficiencies.

Overall, the station-level analysis collectively validates the network-scale statistics while revealing a systematic spatial dependence of fusion accuracy linked to underlying landscape characteristics—a pattern that remains consistent across different soil layers. This site-based spatial evaluation thus usefully complements the network-level validation and further confirms that the DF-based point–surface fusion framework can robustly enhance both the accuracy and spatial reliability of the downscaled SM across multiple soil depths.

To evaluate the station-level generalization capability under spatially independent conditions, a leave-location-out valida-

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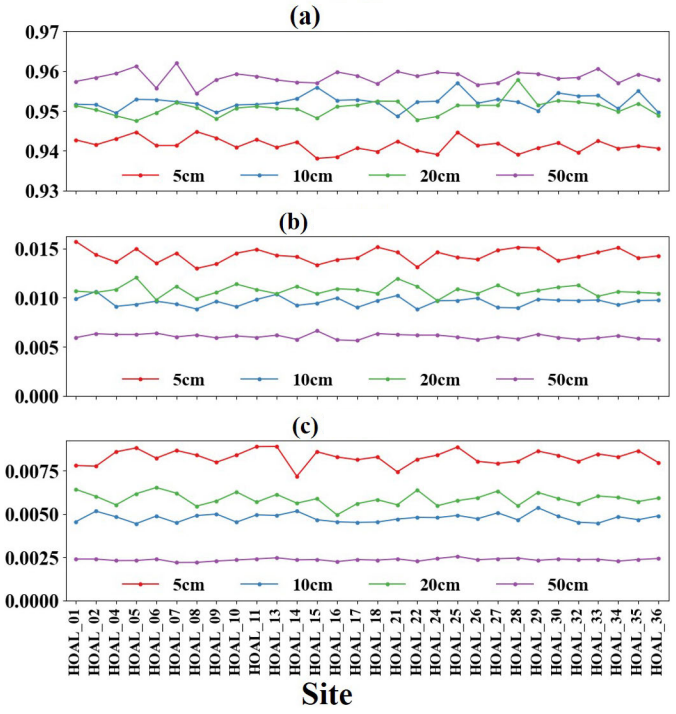


Fig. 7. Leave-location-out validation of the fused multilayer SM. (a)–(c) Station-wise PCC, RMSE, and MAE. Each line represents station-wise performance under spatially independent validation at four soil depths (5, 10, 20, and 50 cm).

tion was conducted using in situ observations. All samples from one station were completely withheld during model training and were used exclusively for validation, ensuring strict spatial independence between training and testing data. Fig. 7 presents the leave-location-out validation result of the fused multilayer SM. The fused SM maintained robust agreement with in situ measurements across all four depths. The mean PCC values reached 0.941, 0.952, 0.951, and 0.958 at depths of 5, 10, 20, and 50 cm, respectively, showing an approximately 3%–5% reduction in correlation strength compared with the nonspatially independent evaluation. Correspondingly, the RMSE ranged from approximately 0.014 $m^3 \cdot m^{-3}$ at 5 cm to 0.005 $m^3 \cdot m^{-3}$ at 50 cm, while MAE values remained below 0.010 $m^3 \cdot m^{-3}$ for all depths, accompanied by moderate increases. This performance degradation is physically reasonable, as the Leave-location-out validation strategy explicitly removes local station information from the training process. Nevertheless, the consistently high PCC values and low error magnitudes demonstrate that the proposed downscaling and fusion approach generalizes well beyond instrumented locations and does not rely on site-specific overfitting.

B. Drought Evaluation and Assessment of DRDI, SMADI and SWDI

The drought monitoring performance of the DRDI proposed in this study was evaluated by calculating daily DRDI values for two crops, winter wheat and summer corn (abbreviated as DRDI-W and DRDI-C, respectively), using the previous high-resolution and multilayer SM data. We also computed the average DRDI during their three important development

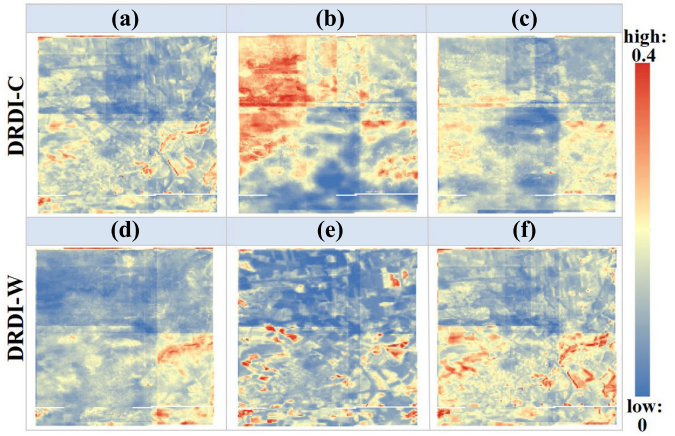


Fig. 8. Spatial distribution of average DRDI of winter wheat and summer corn during three important development stages. (a)–(c) DRDI-C of summer corn at jointing, silking, and maturity stages. (d)–(f) DRDI-W of winter wheat at growing, flowering, and maturity stages.

stages, and the spatial distribution is shown in Fig. 8. Overall, neither wheat nor corn encountered serious drought threats throughout their development, with DRDI ranging between 0 and 0.4, indicating a wet or slightly drought state. Winter wheat had higher water demands, and its maturity stage coincided with the jointing stage of summer corn (May–June). However, drought during the maturity stage of winter wheat was more severe than during the jointing stage of summer corn, due to the more developed root system of winter wheat at the maturity stage, while the SM in the deeper root zone was insufficient. The silking stage of summer corn (July to August) experienced the most severe drought, as the root-zone SM is the lowest level throughout the year. The growing and flowering stages of winter wheat occur in winter and spring, and the root-zone SM during this period is the highest in a year, resulting in a relatively rare occurrence of drought. These findings demonstrate that, compared with individual environmental conditions, the DRDI exhibits a superior capacity in ascertaining drought.

For comparative evaluation, DRDI was further analyzed against two established SM drought indices, SMADI and SWDI, using three representative stations, HOAL-08, HOAL-17, and HOAL-21, situated in HOAL. The daily DRDI, SMADI, and SWDI during 2019 were calculated, and the temporal variations were plotted in Figs. 9 and 10.

Fig. 9 presents the temporal series of DRDI-W, DRDI-C, and SMADI. The temporal variation trends of these three indices maintain a remarkable consistency, fluctuating between 0 and 0.6 and exhibiting distinct seasonality. During the majority of summer, the indices were above 0.2, indicating slight drought, consistent with SM distribution. In general, DRDI value was slightly lower than SMADI during most of the time, which is attributed to the inclusion of root-zone SM in DRDI, allowing deeper soil water availability to be considered. SMADI, in contrast, primarily considers surface SM. However, during periods of increased drought stress (when index values exceed approximately 0.3), DRDI shows a stronger response than SMADI in HOAL-17, reflecting

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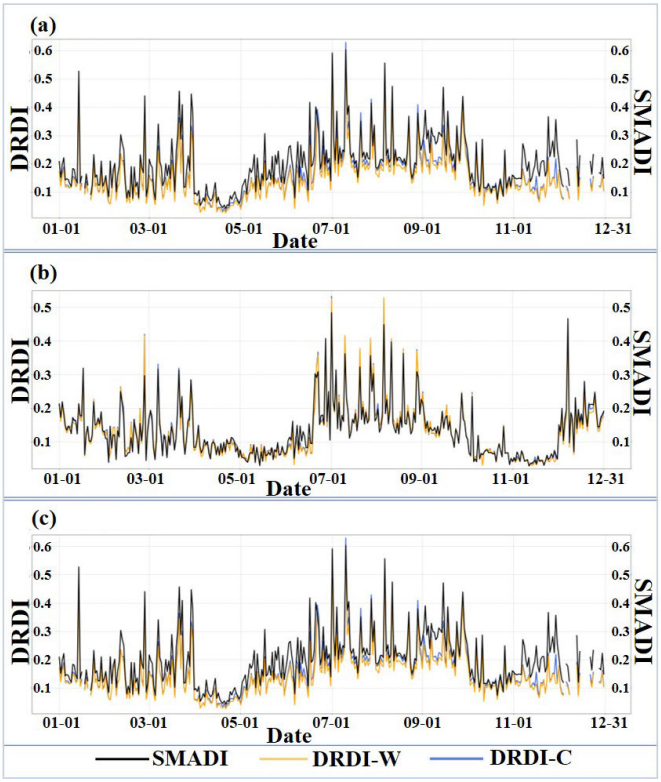


Fig. 9. Temporal variation of daily SMADI, DRDI-W and DRDI-C during 2019. (a)–(c) Temporal variation of stations HOAL-08, HOAL-17, and HOAL-21. The black line represents SMADI. The black, orange, and blue lines represent SMADI, DRDI-W, and DRDI-C, respectively.

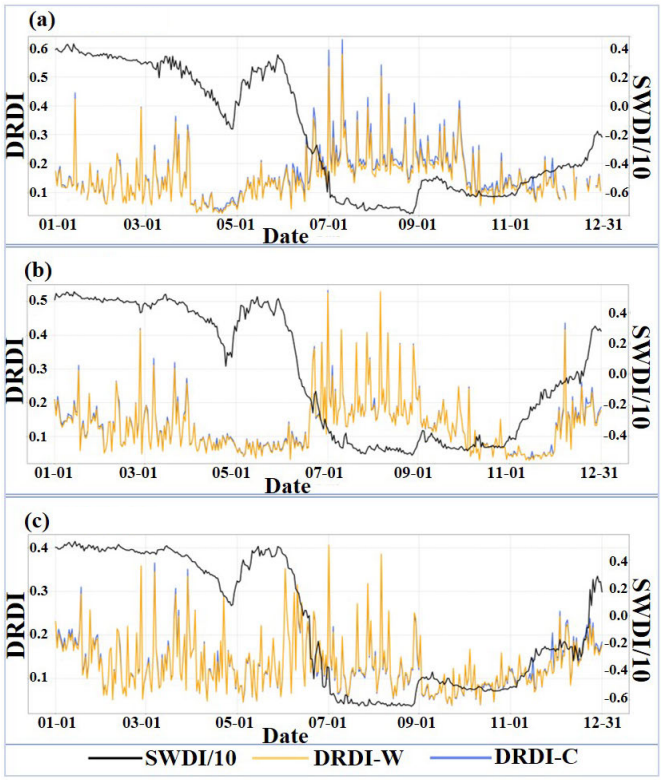


Fig. 10. Temporal variation of daily SWDI, DRDI-W, and DRDI-C during 2019. (a)–(c) Temporal variation of stations HOAL-08, HOAL-17, and HOAL-21. The black line represents SWDI. The black, orange, and blue lines represent SWDI, DRDI-W, and DRDI-C, respectively.

enhanced sensitivity to root-zone water deficits under abnormal agricultural drought conditions. This comparison is intended to demonstrate the improvement of DRDI over its predecessor in agricultural drought representation. DRDI is expected to be more physically meaningful due to the inclusion of root-zone processes and crop-specific water uptake mechanisms.

Fig. 10 presents the temporal series of DRDI-W, DRDI-C, and SWDI. DRDI and SWDI exhibited opposite trends in temporal variation because of their fundamentally different physical definitions. SWDI represents a soil water deficit relative to field capacity, and it typically takes negative values, where more negative values indicate more severe drought conditions. In contrast, higher values of DRDI indicate stronger drought stress. Therefore, as drought severity increases, SWDI decreases (becomes more negative), while DRDI increases, leading to opposite monotonic behaviors in their temporal evolution. From the perspective of a single station, during the summer drought, DRDI at HOAL-21 was lower than that at HOAL-08 and HOAL-17, while SWDI at HOAL-21 remained nearly equivalent to or even surpassed that at HOAL-17. These results demonstrate that DRDI is more capable of reflecting the agricultural droughts with severe intensities occurring in summer compared with SWDI. Both DRDI and SWDI are physically consistent within their respective frameworks, but represent different aspects of drought. This comparison is not intended as a direct performance ranking. Instead, it is used to evaluate the temporal consistency and the physical response differences between a crop-oriented drought index (DRDI) and a hydrological soil water deficit index (SWDI).

To further evaluate temporal behavior and reliability, 95% confidence intervals (CIs) were derived for each index using a bootstrap resampling method during the critical drought period (June to August in 2019). Repeated resampling of model inputs allowed quantification of temporal uncertainty due to input variability and model sensitivity.

Fig. 11 presents the time series and corresponding CIs of DRDI-W, DRDI-C, and SMADI at HOAL-17. Fig. 11(a)–(c) shows the CIs of only one index per panel, with other indices plotted as reference lines. Fig. 11(d) combines all indices for direct comparison. The results indicate a strong consistency between DRDI and SMADI, but both DRDI-W and DRDI-C generally exhibit narrower CIs than SMADI, indicating greater temporal stability and reduced sensitivity to input perturbations. During major drought events (mid-July and early August), DRDI maintains more compact uncertainty bounds, suggesting better robustness. Moreover, Fig. 11(d) further highlights similarities and distinctions among indices. While DRDI-W and DRDI-C share significant overlap, SMADI diverges in several periods, implying that DRDI captures additional information and dynamics not reflected by SMADI alone.

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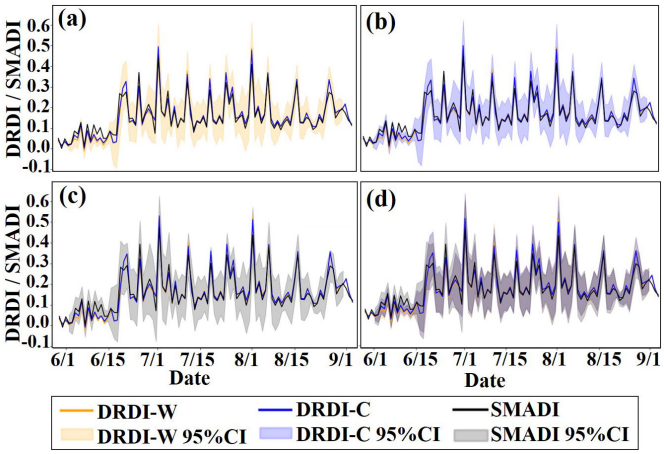


Fig. 11. Temporal variability and uncertainty analysis of SMADI, DRDI-W, and DRDI-C during June to August 2019. The black, orange, and blue lines represent SMADI, DRDI-W, and DRDI-C, respectively. The shaded areas represent the respective CIs of each index. (a)–(c) Respective CIs of DRDI-W, DRDI-C, and SMADI individually, and (d) overlays all indices and their CIs.

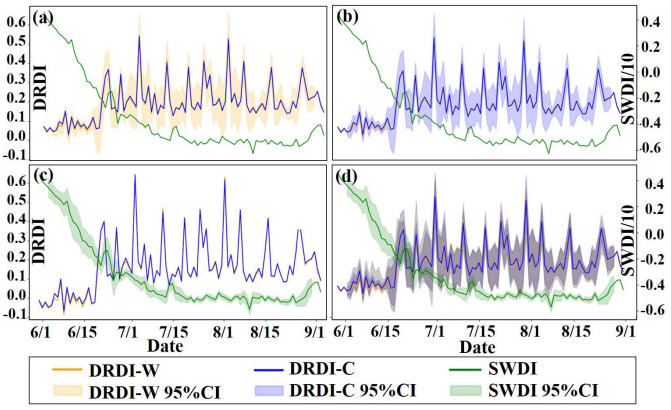


Fig. 12. Temporal variability and uncertainty analysis of SWDI, DRDI-W, and DRDI-C during June to August 2019. The green, orange, and blue lines represent SWDI, DRDI-W, and DRDI-C, respectively. The shaded areas represent the respective CIs of each index. (a)–(c) Respective CIs of DRDI-W, DRDI-C, and SWDI individually, and (d) overlays all indices and their CIs.

Similarly, Fig. 12 presents the time series and CIs of DRDI-W, DRDI-C, and SWDI. The DRDI-W and DRDI-C exhibit more pronounced fluctuations compared to SWDI. Especially in the mid-to-late drought period after July, DRDI shows frequent spikes and drops, while SWDI presents a smoother trend. This difference highlights their distinct response mechanisms. DRDI incorporates high-resolution variables such as multilayer SM, vegetation indices, and dynamic root depth, making it more responsive to short-term meteorological variability. In contrast, SWDI is based on a simpler formulation of soil water deficit, focusing on accumulated moisture stress, and thus responds more slowly to rapid environmental changes. The higher variability of DRDI reflects its design to capture short-term drought dynamics rather than inconsistency. Overall, this uncertainty-based analysis provides further evidence for the robustness of the proposed DRDI model, supporting its suitability for high-resolution, field-scale drought monitoring.

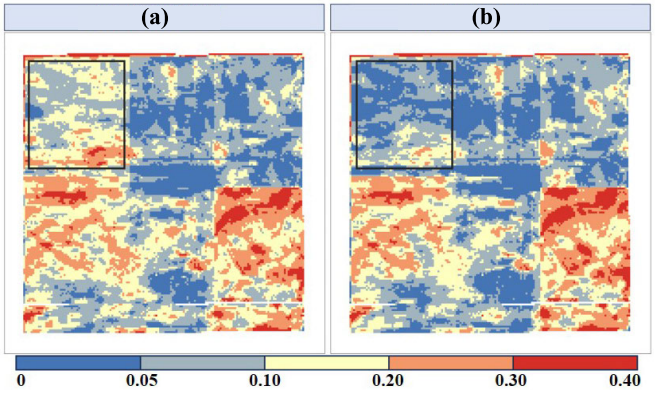


Fig. 13. Spatial distribution of (a) DRDI-C and (b) DRDI-W at the annual scale in 2019.

To assess the agricultural drought situation in 2019, annual-scale DRDI-C and DRDI-W values were calculated in the HOAL region. The spatial distributions of DRDI-C and DRDI-W are shown in Fig. 13. It can be seen that the HOAL suffered relatively less drought, with the DRDI values fluctuating between 0 and 0.4, indicating that the region was either in a wet and drought-free state or in a slight drought state. The central and northern parts of the HOAL were in a relatively wet state, while slight agricultural drought occurred in the southeastern and southwestern regions. The spatial distributions of DRDI-C and DRDI-W are generally similar. However, the values of DRDI-C are significantly higher than those of DRDI-W in the northwestern region. This difference may be related to crop-specific root characteristics and seasonal growing conditions. Winter wheat typically develops a deeper and more uniform root system under cooler conditions, whereas summer corn grows during warmer periods with higher evaporative demand and relies more on near-surface SM [60], [61], [62]. Given the typical winter wheat–summer corn rotation and the temperate humid climate in the HOAL catchment, these differences in root distribution and water use strategies can lead to distinct drought responses. Consequently, more water should be irrigated during the cultivation of summer corn than during that of winter wheat to fulfill the growth demands of the crops.

C. Validation of the Proposed DRDI

To validate the drought monitoring performance of DRDI proposed in this study, we calculated the decreased values of gross primary productivity (GPP) and leaf area index (LAI) compared to the previous time period, denoted as GPP- and LAI-, respectively. DRDI represents the instantaneous intensity of agricultural drought stress experienced by crops, rather than its temporal rate of change. Similarly, GPP- and LAI- represent the magnitude of reduction in vegetation productivity and canopy structure relative to a reference condition, which directly reflects the impact of current drought stress on vegetation. They are all state variables describing stress intensity, and therefore we use GPP- and LAI- as proxies for drought-induced impacts on crop growth.

Fig. 14 presents the validation result during 2019. The correlations between GPP-, LAI-, and DRDI were investigated

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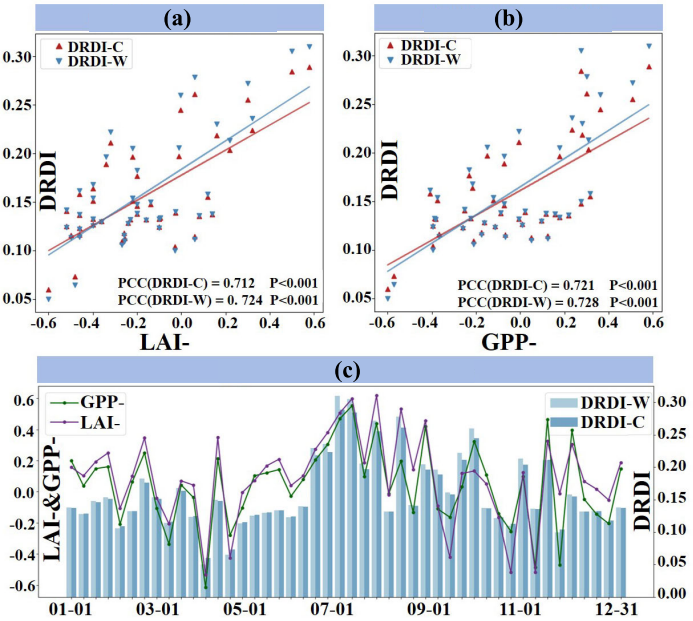


Fig. 14. Validation of DRDI during 2019. (a) and (b) Scatter plots of LAI- and GPP- with DRDI. The red regular triangles represent DRDI-C and blue inverted triangles represent DRDI-W. (c) Temporal variation of daily DRDI with LAI- and GPP in station HOAL-08 during 2019.

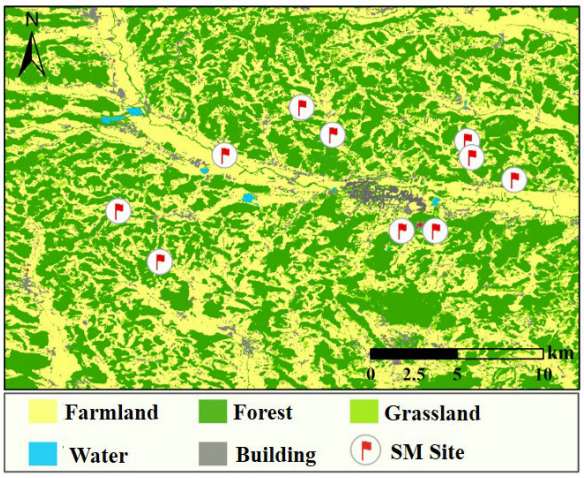


Fig. 15. Geographical location and land cover distribution of the Wagna.

using scatter plots, as shown in Fig. 14(a) and (b). The PCC values of DRDI with LAI- and GPP- all exceed 0.712, and the Diebold–Mariano significance tests yielded probability values (p-values) all below 0.001, all indicating relatively high correlations. The temporal variation of GPP-, LAI-, and DRDI for the station HOAL-08 is shown in Fig. 14(c). Overall, GPP- and LAI- exhibit similar temporal variation trends, which are consistent with that of DRDI-W and DRDI-C. Given that both GPP and LAI are indicators characterizing the stability of crop growth status, GPP- and LAI- will also exhibit favorable temporal consistency. Consequently, it can be demonstrated that the DRDI, an agricultural drought index of SM that can be dynamically adapted to the development of crops, can effectively reflect the negative impact of drought on crop growth.

D. Independent Validation of Spatiotemporal Generalization

Validation limited to a single region (HOAL) and a single calendar year (2019) may lead to extremely high PCC (> 0.981) and low RMSE (< 0.012 $m^3 \cdot m^{-3}$). To more rigorously assess the robustness of the proposed SM fusion framework and the derived DRDI beyond the training footprint, independent validation experiments focusing on regional-scale spatial transferability and interannual temporal robustness were conducted.

For the spatial generalization test, the Wagna catchment in Styria, southeastern Austria, was chosen as an independent study area (Fig. 15). This region is approximately 150 km from the HOAL and characterized by farmland-dominated land cover (mainly corn, wheat, and grapes), with patches of forest and grassland contributing to agricultural adaptability. Wagna is part of Austria's national hydrological and ecological monitoring network, providing multiple in situ SM observation sites. Although Wagna and HOAL share similar climatic conditions, their differences in land cover and soil properties make Wagna a suitable testbed for evaluating regional-scale spatial transferability.

The SM fusion model was trained exclusively using HOAL data from 2019 and subsequently applied, without retraining, to generate 30-m resolution multilayer SM estimates over the Wagna catchment for the same year. These predictions were then evaluated against independent in situ SM observations. As shown in Fig. 16, the fused SM exhibits consistent agreement with in situ measurements across all four depths (5, 10, 20, and 50 cm), with PCC values exceeding 0.780 and RMSE remaining below 0.032 $m^3 \cdot m^{-3}$. Compared to the in-sample evaluation in HOAL, PCC values decrease by approximately 18.9%, reflecting the expected performance degradation under spatial extrapolation. Importantly, the model maintains stable and physically reasonable accuracy levels, indicating that the high performance observed in HOAL is not solely attributable to spatial leakage or overfitting.

To evaluate temporal robustness, an independent validation was conducted using data from 2018, a climatically wetter year relative to the training year (2019). The model trained on HOAL 2019 data was directly applied to predict multilayer SM for HOAL in 2018, and the results were compared with in situ observations. The scatter density plots in Fig. 16 demonstrate that PCC values at depths of 5, 10, 20, and 50 cm all exceed 0.874, while RMSE remains below 0.027 $m^3 \cdot m^{-3}$. Relative to the training year, the average PCC reduction is approximately 9.3%, which is substantially smaller than the decrease observed in the spatial extrapolation experiment. This result suggests that the proposed fusion framework exhibits strong interannual robustness under varying hydroclimatic conditions.

In addition, the independently derived SM data we also used to calculate the DRDI to further quantify its correlation with vegetation growth indicators GPP- and LAI- in the Wagna region in 2018 year. The corresponding scatter plots are presented in Fig. 17. Fig. 17(a) and (b) displays the results of the spatial extrapolation test. Compared with the original training region, the PCC only slightly declined by 5.399%, indicating

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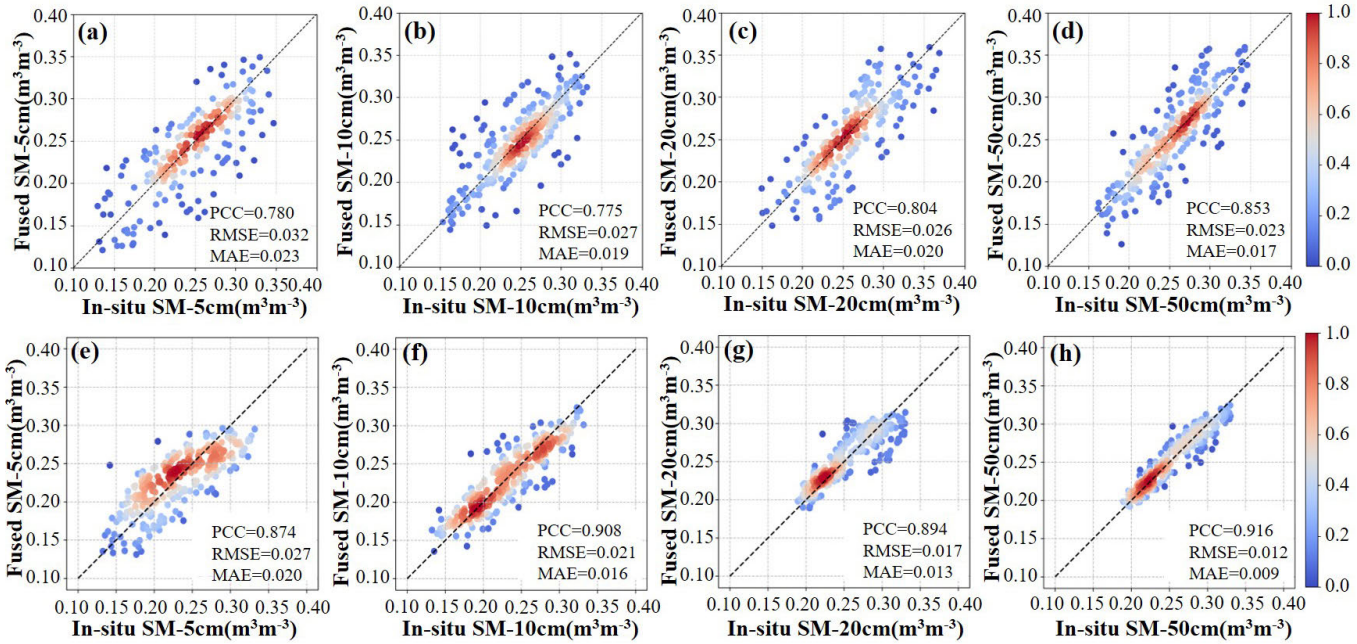


Fig. 16. Scatter plots and accuracy between fused SM and in situ SM at depths of 5, 10, 20, and 50 cm, respectively, in spatiotemporal extrapolation test. (a)–(d) Show the scatter plots between fused SM and in-situ SM at depths of 5 cm, 10 cm, 20 cm, and 50 cm respectively in spatial extrapolation test and (e)–(h) show the scatter plots between fused SM and in-situ SM at depths of 5 cm, 10 cm, 20 cm, and 50 cm respectively in temporal extrapolation test.

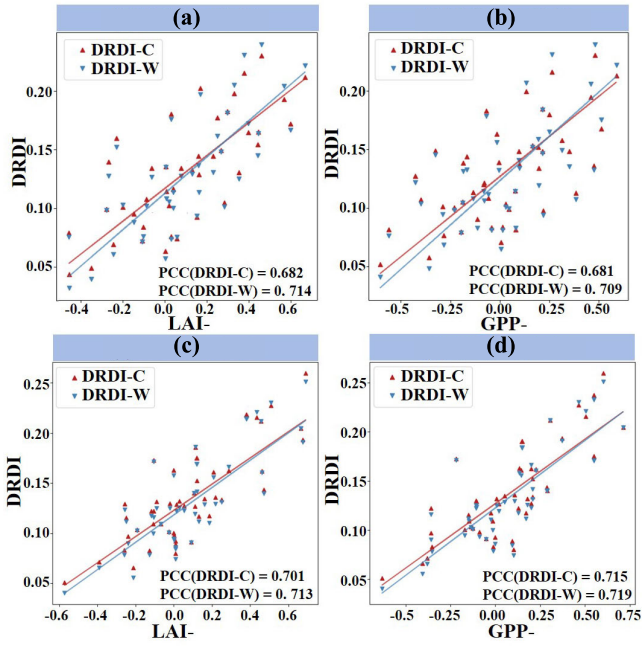


Fig. 17. Scatter plots of LAI- and GPP- with DRDI in spatiotemporal extrapolation test. (a) and (b) Scatter plots in spatial extrapolation test. (c) and (d) Scatter plots in temporal extrapolation test. The red regular triangles represent DRDI-C and blue inverted triangles represent DRDI-W.

that the DRDI still possesses reliable drought monitoring capability in unseen spatial regions. The scatter plots of spatial extrapolation are shown in Fig. 17(c) and (d). Compared with the original training year, the PCC decreased by only 3.294% on average, a smaller drop than that observed in the temporal extrapolation test. These results indicate that, although the absolute performance of SM estimation decreases under independent validation, the derived DRDI demonstrates enhanced robustness at the drought-index level, benefiting from the integrated use of multilayer SM, vegetation condition, temperature stress, and crop-specific root-depth weighting.

Overall, the independent spatial and temporal validation experiments demonstrate that the proposed multilayer SM fusion method and the corresponding DRDI framework possess robust regional-scale spatial transferability and inter-annual temporal robustness. While these results do not imply universal applicability across all climatic zones or management regimes, they provide strong evidence that the framework can be reliably transferred beyond the training region and year, supporting its potential for broader agricultural drought monitoring applications.

V. Discussion

A. Scale Assumption and Its Impact on Downscaling

The proposed downscaling framework establishes the relationship between SM and environmental variables at the coarse spatial resolution (9 km) and subsequently applies this relationship to fine-resolution (30 m) inputs. This process implicitly assumes that the relationship learned at the coarse scale is transferable across spatial scales and is commonly adopted in statistical downscaling approaches [63], [64]. However, it may introduce uncertainties due to the presence of substantial land surface heterogeneity within coarse-resolution pixels.

In this study, we mitigate this issue by explicitly incorporating high-resolution environmental variables during the inference stage. Specifically, variables including LST, NDVI, albedo, precipitation, and DEM are used at 30-m resolution.

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TABLE V
COMPARISON OF SM DOWNSCALING RELATED STUDIES

Studies	Methods	Spatial Resolution	Temporal Resolution	SM Layers	Spatial Seamless
[65]	CNN	1 km	1day	SSM	no
[41]	ESTARFM, DBN	30 m	1day	SSM	yes
[66]	CDF Matching	0.25°	1day	SSM, RZSM	no
[67]	XGBoost	1 km	1day	SSM	yes
[27]	Exponential filter	1 km	1day	RZSM	yes
[30]	XGBoost	1 km	1day	5cm, 10cm, 20cm, 50cm, 100cm	no
Our study	**ESTARFM, ERF, DF**	**30 m**	**1day**	**5cm, 10cm, 20cm, 50cm**	**yes**

These variables represent key controlling factors of SM variability, such as surface energy balance, vegetation conditions, topographic effects, and water input. By providing fine-scale spatial information, these predictors enable the model to capture intrapixel heterogeneity when generating high-resolution SM estimates. All input variables are first aggregated to 9-km resolution during model training to ensure consistency with the SMAP L4 product. After the model is trained, it is then applied to the corresponding variables at their native 30-m resolution. This workflow follows the general paradigm of statistical downscaling, where coarse-scale relationships are combined with high-resolution predictors to reconstruct fine-scale spatial patterns.

The impact of this scale assumption on the reliability of the results is evaluated using in situ observations. The validation results demonstrate that the downscaled SM shows good agreement with ground measurements across multiple depths. These results indicate that, despite the scale assumption, the proposed method is capable of effectively capturing spatial variability and producing reliable SM estimates.

Nevertheless, uncertainties may still exist in regions with strong subpixel heterogeneity or processes with highly nonlinear behavior that cannot be fully represented by the selected predictors. Future work could further explore scale-aware or multiscale modeling strategies to explicitly account for scale-dependent relationships and improve the robustness of downscaling results.

B. Causes of Blockiness and Spatial Discontinuities in Fused SM

A visible block-like pattern was observed in the fused SM in the HOAL region. This spatial pattern primarily originates from the inherent characteristics of the input datasets and the structural properties of the fusion model. The core variable used in the fusion framework—SMAP—has a relatively coarse spatial resolution of 9 km, which means that only four SMAP pixels cover the entire HOAL area. As a result, the fused SM field tends to inherit this coarse-scale spatial structure, forming four visually distinct regions corresponding to the SMAP pixel boundaries. In addition, SM exhibits extremely strong spatial heterogeneity due to variations in soil texture, vegetation, and microtopography. When geographic coordinates (latitude and longitude) are incorporated as auxiliary predictors to capture this heterogeneity, the model tends to enhance localized spatial differences, which further contribute to the block-like spatial segmentation pattern. Moreover, minor discrepancies in spatial alignment and resampling between multisource datasets with differing resolutions may accentuate these transitions at the pixel boundaries. Overall, this block-like pattern is physically interpretable and consistent with the spatial structure of the SMAP data and the intrinsic heterogeneity of SM. It does not indicate an artifact or flaw in the fusion process. In future work, an adaptive spatial smoothing strategy will be explored to further refine spatial continuity while preserving the authentic variability of SM across different landscape units.

C. Role of High-Resolution SM in Agricultural Drought Monitoring

Most existing official SM products, such as the 9-km SMAP and 0.1° ERA5-Land, have relatively coarse spatial resolution, which leads to insufficient number of pixels to capture detailed local environmental characteristics, particularly evident in agricultural cultivation regions with relatively small areas. In this study, we addressed these limitations by downscaling the originally coarse-resolution SM datasets through the integration of high-resolution remote sensing data and in situ measurements. Table V compares the SM data generated in our study with existing downscaled SM data generated by other studies in terms of methods, spatial resolution, temporal resolution, and SM layers. It is clear that only our study achieved the acquisition of a multilayer SM dataset with a 30-m high spatial resolution among all the comparative studies.

The superiority of our generated 30-m resolution multilayer SM dataset was further evaluated through a comparative experiment. A 1-km resolution SM product generated using

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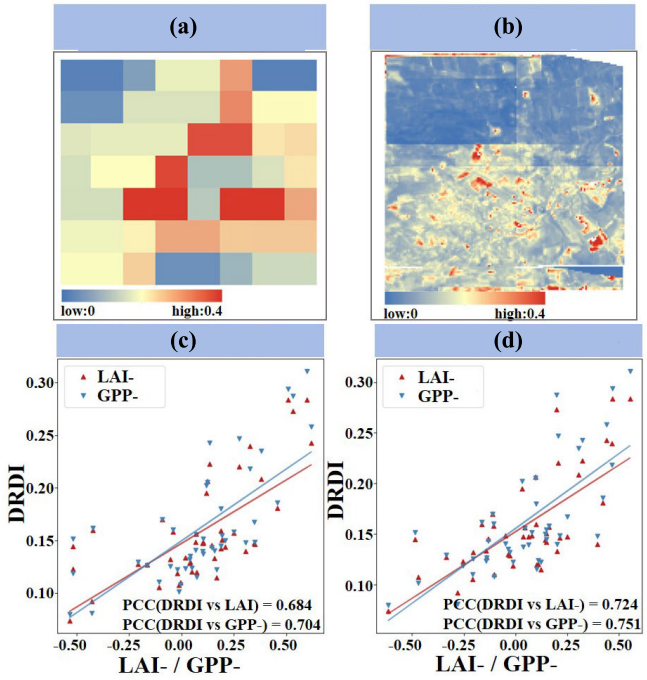


Fig. 18. Spatial distribution of DRDI by different SM and scatter plots between LAI- and GPP- with DRDI. (a) DRDI calculated by 1-km SM product obtained from [30] in Table V. (b) DRDI calculated using the fused 30-m SM generated by this study. (c) Scatter plots between 1-km DRDI and LAI-/GPP-. (d) Scatter plots between 30-m DRDI and LAI-/GPP-.

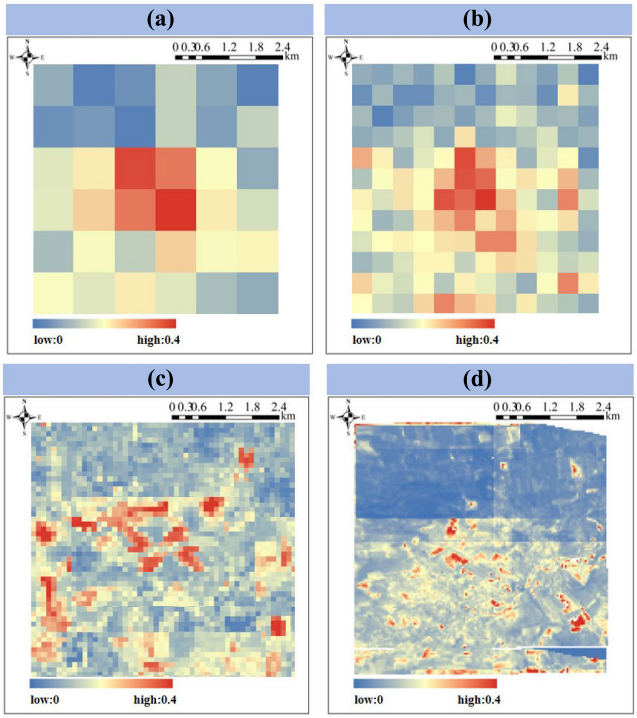


Fig. 19. Spatial distribution of DRDI by SM data at different spatial resolutions. (a) DRDI by 1-km SM. (b) DRDI by 500-m SM. (c) DRDI by 100-m SM. (d) DRDI by fused 30-m SM.

the method based on [30] in Table V served as the comparison product. Then DRDI were computed for both datasets. The spatial distribution of the two DRDIs along with the scatter plot against GPP- and LAI- are shown in Fig. 18. Fig. 18(a) and (b) displays the summer corn DRDI on March 22, 2019. The spatial patterns are not perfectly aligned. This phenomenon is mainly attributed to the differences in spatial resolution, inevitably leading to variations in the subsequently calculated DRDI. The DRDI at 1-km DRDI exhibited pronounced spatial smoothing, obscuring details of drought intensity changes and limiting the ability to reveal drought heterogeneity at local scales. In contrast, the 30-m DRDI demonstrated superior representation of spatial heterogeneity, delineating differences in drought severity between agricultural plots and effectively identifying microscale drought hotspots. This reflects spatial variability in SM driven by topography, soil type, and crop distribution patterns, enhancing its value for precision farm management applications.

Fig. 18(c) and (d) presents scatter plots and PCC between two DRDIs with GPP-/LAI- in 2019 to quantify the drought index's responsiveness to crop yield correlations. Results show that DRDI based on our 30-m SM data achieved higher correlations, and the average PCC with GPP- and LAI- reached 0.751 and 0.724, respectively. This performance is much higher than the 1-km DRDI. Overall, compared to the 1-km SM, our 30-m SM-driven drought index improved characterization of crop response capacity by 6.67% (GPP-) and 5.85% (LAI-). These results validated the significant advantages of high-resolution and multilayer SM data generated in this study for agricultural drought monitoring. Specifically, at the field scale, it captured crop root-zone SM dynamics with greater granularity, thereby enhancing the drought index's capacity to characterize crop physiological responses.

D. Sensitivity of DRDI to Spatial Resolution

Due to unaligned spatial patterns caused by different spatial resolution data, we assessed the stability of the DRDI across different spatial scales by analyzing its sensitivity to the spatial resolution of input data. Specifically, we generated additional multilayer SM datasets with coarser resolutions of 100 m, 500 m, and 1 km by altering the spatial resolution of the input data used in the proposed fusion framework. Subsequently, DRDI corresponding to each resolution was calculated. As illustrated in Fig. 19, the DRDI maintained highly consistent spatial drought patterns across resolutions ranging from 1 km to 30 m. This indicates the robustness of the DRDI framework, ensuring the comparability of DRDI results derived from heterogeneous data sources. Moreover, compared with the DRDI at 1 km, 500 m, and 100 m, the 30-m high-resolution DRDI [Fig. 19(d)] exhibited superior capability in capturing fine-scale drought details. It clearly delineated intrafield variations and differences in drought conditions across areas with distinct vegetation covers—heterogeneous information that was substantially averaged out and unidentifiable at 1 km [Fig. 19(a)] and 500 m [Fig. 19(b)]. Although the 100-m resolution DRDI [Fig. 19(c)] displayed partial contours, it still fell far short of the level of detail provided at 30 m.

In summary, these findings confirm the stability of the proposed DRDI framework across spatial scales, while also highlighting the necessity of the 30-m high-resolution SM

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TABLE VI
COMPARISON OF AGRICULTURAL DROUGHT INDICES RELATED STUDIES

Studies	Indices	Constituent Factor	Crop Water Demand Consideration	Crop Root Dynamic Growth Consideration
[14]	PDSI	Precipitation, PET, SM	no	no
[68]	SPEI	Precipitation, PET	no	no
[69]	CMI	Precipitation, SM, ET, PET	yes	no
[16]	ARID	Precipitation, SM, PET	yes	no
[18]	SMADI	SM, NDVI, LST	no	no
[70]	MSWDI	SM	yes	no
[17]	MSDI	Precipitation, SM	no	no
[71]	CADI	SM, ET, Precipitation	no	no
Our study	**DRDI**	**SM, NDVI, LST, RLD**	**yes**	**yes**

dataset and corresponding agricultural drought index for precision agricultural drought monitoring, which provides unprecedented field-level drought detail, offering robust scientific support for differentiated drought management strategies.

E. Advancement of Considering Crop Root Development in Agricultural Drought Indices

Table VI compares the DRDI proposed in our study with other agricultural drought indices from existing research. Most agricultural drought indices only consider SM and precipitation, with a few taking into account the water requirement of the crop. Only the DRDI considers the changes in crop water demand at different root depths during different growth stages. By introducing the RLD, DRDI captures the root distribution characteristics of different crops throughout their developmental stages. Agricultural drought indices that incorporate such crop physiological structure information (root distribution) are capable of precisely reflecting the actual drought situations that crops encounter at different stages. Consequently, they can offer precise guidance for irrigation, preventing the waste of water resources during drought. Our work enables drought-resistant measures to be more targeted and scientific.

It is also acknowledged that obtaining accurate RLD data involves inherent uncertainties due to the challenges of direct field measurements and limited dataset applicability. To quantify the sensitivity of the proposed DRDI index to RLD uncertainty, we conducted a stratified RLD perturbation experiment. Independent perturbations of ±5%, ±10%, and ±20% were applied to the RLD values at each of the four soil layers (5, 10, 20, and 50 cm), generating a total of 24 distinct perturbation scenarios. For each scenario, we calculated the PCC between DRDI and GPP- and LAI- as shown in Fig. 20.

DRDI exhibited notable depth dependence, directional asymmetry, and crop-specific responses to RLD perturbations.

More specifically, DRDI sensitivity decreases with increasing soil depth. Perturbations in the surface layer at 5 cm lead to the most notable changes; a −20% RLD perturbation leads to an increase in the PCC between DRDI-W and LAI- to 0.737 (ΔPCC of + 0.015). In contrast, perturbation at 50 cm produced changes of + 0.004, indicating strong robustness. Meanwhile, DRDI exhibited a markedly asymmetric response to RLD perturbations: negative perturbations (reductions in RLD) generally improved correlations, while positive

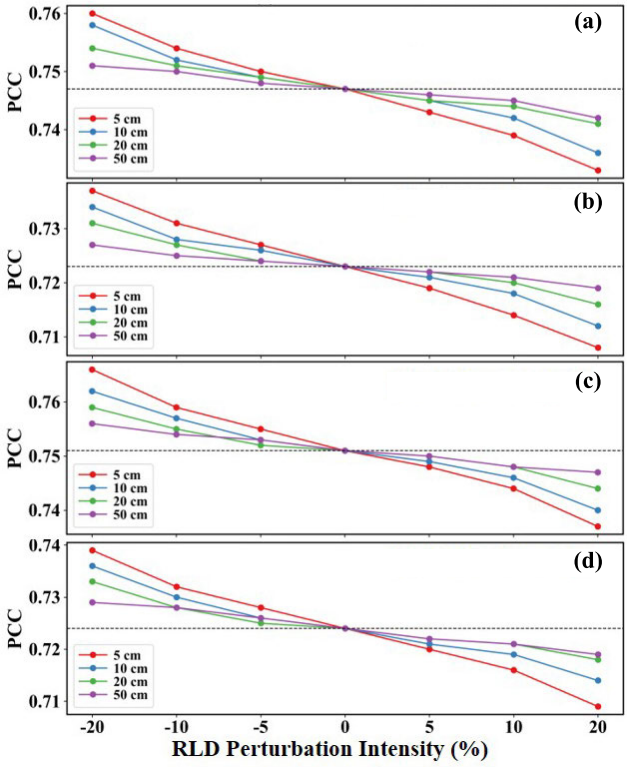


Fig. 20. Sensitivity of RLD uncertainty to DRDI stability. (a) PCC of DRDI-W and GPP- under perturbation. (b) PCC of DRDI-W and LAI-. (c) PCC of DRDI-C and GPP-. (d) PCC of DRDI-C and LAI-.

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perturbations reduced them. This asymmetry stems from the intrinsic design of the DRDI calculation, where RLD functions as an amplification factor for moisture stress signals. When RLD is overestimated, it can exaggerate drought conditions, leading to a reduction in correlation. Conversely, underestimation of RLD enhances the index's accuracy. In addition, crop-specific responses were evident: corn showed slightly higher sensitivity (average $|\Delta PCC|$ of 0.0056) than wheat (average $|\Delta PCC|$ of 0.0053), particularly in deep layers. Corn was also more sensitive to deep-layer perturbations, while wheat showed stronger responses to surface-layer changes. This difference likely arises from their distinct root system architectures. Corn has a more deeply developed system, which makes corn more sensitive to changes in RLD in deeper layers. In addition, GPP- showed higher sensitivity to RLD perturbations than LAI-. It can be attributed to the fact that GPP- reflects the photosynthetic carbon assimilation process, which responds more immediately to water stress. In contrast, LAI- typically exhibits a delayed response.

Despite the demonstrated robustness of DRDI to moderate RLD perturbations, several limitations related to transferability should be acknowledged. In the current implementation, RLD values are adopted from prior studies and treated as static, crop-specific parameters. As a result, spatial variability associated with soil texture, nutrient availability, irrigation practices, cultivar traits, and climatic conditions is not explicitly represented. Consequently, while the proposed DRDI framework is transferable in principle, its direct application to other regions or cropping systems may require adjustment of the RLD weighting profiles. In environments where rooting depth or vertical root distribution deviates substantially from the literature-based assumptions used here, unmodified RLD parameters may introduce bias in the depth-weighted SM aggregation and drought intensity estimation.

In summary, DRDI demonstrated only mild sensitivity to RLD uncertainty and remained particularly robust under deep-layer perturbations. Despite the inherent uncertainties in RLD estimation, the proposed DRDI retains strong drought monitoring capabilities under the tested conditions. Importantly, the DRDI framework is designed to be modular and adaptable. Since the depth-wise weighting parameters are derived directly from crop-specific RLD distributions, they can be readily replaced or recalibrated when regionally representative RLD information is available. This flexibility allows DRDI to accommodate crops with contrasting rooting strategies, ranging from shallow-rooted species (e.g., rice and vegetables) to deep-rooted crops (e.g., maize, sugarcane, and fruit trees). Therefore, while site-specific parameterization is recommended for cross-regional applications, the overall DRDI formulation provides a scalable and extensible basis for crop-aware drought assessment across diverse agricultural contexts.

VI. Conclusion

Although many SM-based agricultural drought indices have been proposed, most of them neglect crop water requirements at different root-zone depths and do not account for crop-specific developmental stages. The key unsolved issue is the lack of high-resolution, multilayer SM datasets at the field scale. Considering that, we proposed a high-resolution, multilayer SM data fusion framework and constructed a new agricultural drought index, i.e., DRDI. This new data and drought index accurately capture root-zone SM dynamics and reveal agricultural drought conditions at the field scale. The main conclusions are as follows.

1) The proposed in situ and remote sensing data fusion framework effectively bridges the scale gap between coarse-resolution datasets and field-scale measurements, producing multilayer SM data at daily and 30-m scale. The fused dataset accurately represents SM dynamics across different layers.
2) By incorporating multilayer SM with crop-specific root depth and developmental stage information, DRDI accurately reflects root-zone SM dynamics and captures drought heterogeneity. Compared with existing indices, DRDI demonstrates enhanced sensitivity to severe droughts and shows robust performance across different crop types and agricultural contexts.
3) Application of DRDI in the study area revealed spatial and temporal drought patterns at the field scale in the HOAL. It highlighted variations in drought frequency across land cover types and seasons, providing insights into local drought heterogeneity and crop-specific water stress.

Overall, this study integrates high-resolution, multilayer SM data with crop-specific drought indices, significantly improving the accuracy and reliability of field-scale agricultural drought monitoring. The proposed approach provides a robust framework for precision agricultural water management and informed decision-making enhanced by geoscience techniques and remote sensing datasets.

Code Availability

All analyses were conducted using custom scripts written in Python (version 3.10). The codes are publicly available via the GitHub repository (https://github.com/liuxin1211/DRDI.git).

Data Availability

The fused soil moisture case dataset generated in this study is available at (https://github.com/liuxin1211/DRDI.git).

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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